Module 3 - Basic Arithmetic Operations cont.

Chapter 7- Average

---

1. Find the average of the following set of scores

   $$253, 124, 255, 534, 836, 375, 101, 443, 760$$

   a) 427 b) 413 c) 419 d) 490 e) None of the above

2. Find the average of the following set of scores.

   $$124, 856, 331, 227, 963, 338, 259, 662.$$

   a) 570 b) 660 c) 741 d) 350 e) None of the above

3. What will be the average of even numbers between 11 to 63?

   a) 37.5 b) 47 c) 42 d) 37

4. Find average of all prime numbers between 60 and 90.

   a) 72 b) 74.7 c) 74 d) 73.6

5. Calculate the average of the cubes of first five natural numbers.

   a) 55 b) 65 c) 45 d) 35 e) None of the above

6. The average of 8 numbers is 14. The average of 6 of these numbers is 16. What is the average of the remaining two numbers?

   a) 12 b) 6 c) 8 d) 10 e) None of the above

7. Average age of 5 boy is 16 yr, of which that of 4 boys is 16 yr 3 months. The age of the 5th boy is

   a) 15 yr b) 15 yr 6 months c) 15 yr 4 months d) 15 yr 2 months

8. If the algebraic sum of deviations of 20 observations measured from 23 is 70, mean of these observations would be

   a) 24 b) 25 c) 26 d) 27 e) None of the above

9. Eight consecutive numbers are given. If the average of the two numbers that appear in the middle is 6, then the sum of the eight given numbers is

   a) 36 b) 48 c) 54 d) 64

10. The average of 4 consecutive even numbers is 27. What is the greatest number?

    a) 28 b) 26 c) 32 d) 30

11. A businessman purchased 30 TV sets in ₹300000. If the average cost of 15 TV sets is ₹9000, then find out the average cost of remaining TV sets.

    a) ₹20000 b) ₹12000 c) ₹11000 d) ₹10000 e) None of the above

12. The average age of 30 girls is 13 yr. The average of first 18 girls is 15 yr. Find out the average age of remaining 12 girls.

    a) 12 yr b) 10 yr c) 16 yr d) 10.5 yr e) None of the above

13. The average of four consecutive numbers A, B, C and D is 49.5. What is the product of B and D.

    a) 2499 b) 2352 c) 2450 d) 2550 e) None of the above

14. The average height of the basketball team A is 5 ft 11 inches and that of B is 6 ft 2 inches. There are 20 players in team A and 18 players in team B. The overall average height is

    a) 72.42 inches b) 72 inches c) 70.22 inches d) 70 inches

15. The average of 13 results is 60. If the average of first 7 results is 59 and that of last 7 results is 61, what will be the seventh result?

    a) 90 b) 50 c) 75 d) 60 e) None of the above

16. The average of 50 numbers is 38. If the two numbers 45 and 55 are not considered, what will be the average of remaining numbers?

    a) 36.5 b) 37 c) 37.5 d) 37.52

17. Average weight of 19 men is 74 kg and the average weight of 38 women is 63 kg. What is the average weight (rounded off to the nearest integer) of all the men and the women together?

    a) 59 kg b) 65 kg c) 69 kg d) 67 kg e) 71 kg

18. The average marks of 30 students are 45 but after checking, there are two mistakes found. After adjustment, if a student got 45 more marks and other student got 15 less marks, then what will be the adjusted average?

    a) 45 b) 44 c) 47 d) 46

19. The average of nine numbers is 50. The average of the first five numbers is 54 and that of the last three numbers is 52. Then, the sixth number is

    a) 34 b) 24 c) 44 d) 30

20. The mean marks obtained by seven students in a group is 226. If the marks obtained by six of them are 340, 180, 260, 56, 275 and 307 respectively, find the marks obtained by the seventh student.

    a) 164 b) 226 c) 340 d) Cannot be determined

21. A cricketer has a mean score of 60 runs in 10 innings. Find out how many runs are to be scored in the eleventh innings to raise the mean score to 62?

    a) 80 b) 81 c) 83 d) 82

22. The mean weight of 34 students of a school is 42 kg. If the weight of the teacher be included, the mean rises by 400 g. Find the weight (in kg) of the teacher.

    a) 66 b) 56 c) 55 d) 57

23. The average weight of 21 boys was recorded as 64 kg. If the weight of the 'teacher was added, the average increased by one kg. What was the teacher's weight?

    a) 86 kg b) 64 kg c) 72 kg d) 98 kg e) None of the above

24. The average age of 14 girls and their teacher's age is 15 yr. If teacher's age is excluded, then the average reduced by 1. What is the teacher's age?

    a) 29 yr b) 35 yr c) 32 yr d) 30 yr

25. The average of six observations is 12. The average decreases by 1 when a new observation is included. What is the seventh observation?

    a) 1 b) 3 c) 5 d) 6

26. The average age of a class is 35 yr. 6 new students with an average age of 33 yr joined in that class, there by decreasing the average by half year. The original strength of the class was

    a) 14 b) 18 c) 16 d) 20 e) None of the above

27. The average age of 5 members of a family is 25 yr. If the servant of the family is included, the average age increases by 40%. What is the age of servant?

    a) 85 yr b) 90 yr c) 73 yr d) 55 yr

28. The average age of 4 members of a family is 25 yr. If head of the family is included in this group, then average age increases by 20%. Find out the age of the head.

    a) 45 yr b) 50 yr c) 55 yr d) 60 yr

29. The average weight of 10 women is increased by 1 kg if 2 women of 42 kg and 38 kg are replaced by 2 men. Find the average weight of both the men.

    a) 46 kg b) 45 kg c) 42 kg d) 44 kg e) None of the above

30. The average age of a committee of 11 persons increases by 2 yr when 3 men of 32 yr, 34 yr and 33 yr are replaced by 3 women. What will be the average age of those 3 women?

a) 40 yr b) $41\frac{1}{3}$ yr c) 41 yr d) $40\frac{1}{3}$ yr e) None of the above

31. The average age of 30 women decreases by 3 months if a new person Priyanka is included in place of a 25 yr old woman. Calculate the age of Priyanka.

    a) 17.5 yr b) 20 yr c) 30 yr d) 22 yr e) None of the above

32. Average of ten numbers is 7. If every number is multiplied by 12, what will be the average of new numbers?

    a) 79 b) 92 c) 82 d) 84

33. The average temperature of Delhi for four days in a particular month is 48°C. If the average temperature of second and third days is 34° C and the ratio of the temperatures of first and fourth days is 9 : 11, then find out the temperatures of first and fourth days.

    a) 45.5° C, 49.9° C b) 32.4° C, 39.8° C c) 43.4° C, 68.2° C

    d) 52.4° C, 46.8° C e) None of the above

34. The mean temperature of Monday to Wednesday was 37°C and of Tuesday to Thursday was 34°C. If the temperature on Thursday was 4/5 that of Monday, the temperature of Thursday was

    a) 35.5° C b) 34° C c) 36.5° C d) 36° C

35. The average of 11 numbers is 36. Out of these 11 numbers, the average of 9 numbers is 34. If the ratio of remaining two numbers is 2 : 3, find out the smaller number.

    a) 18 b) 36 c) 54 d) 48 e) None of the above

36. Roshan bought 5 pants at ₹25 each, 10 shirts at ₹50 each and 15 ties at ₹35 each. Find the average price of all the articles.

    a) ₹38.33 b) ₹45 c) ₹60 d) ₹45.33 e) None of the above

37. The average weight of a class of 15 boys and 10 girls is 38.4 kg. If the average weight of the boys is 40 kg, then what is the average weight of the girls?

    a) 36.5 kg b) 35 kg c) 36 kg d) 35.6 kg e) None of the above

38. The mean weight of 150 students in a class is 60 kg. The mean weight of boys is 70 kg and that of girls is 55 kg. What is the number of boys in the class?

    a) 50 b) 60 c) 75 d) 100

39. The average monthly income of 4 earning members of a family is ₹7350. One member passes away and the average monthly income becomes ? 6500. What was the monthly income of the person, who is no more?

    a) ₹6928 b) ₹8200 c) ₹9900 d) ₹12850

40. The average of 5 consecutive odd positive integers is 9. The least one among them is

    a) 5 b) 3 c) 1 d) 7

41. The average of the test scores of a class of m students is 70 and that of n students is 91. When the scores of both the classes are combined, the average is 80. What is $\frac{n}{m}$?

    a) $\frac{11}{10}$ b) $\frac{13}{10}$ c) $\frac{10}{13}$ d) $\frac{10}{11}$

42. If $47a + 47b = 5452$, what is the average of a and b?

    a) 116 b) 23.5 c) 96 d) 58 e) None of the above

43. The average salary per head of all workers of an institution is ₹60. The average salary per head of 12 officers is ₹400. The average salary per head of the rest is ₹56. Then, the total number of workers in the institution is

    a) 1030 b) 1032 c) 1062 d) 1060

44. A car travels a certain distance from town A to town B at the speed of 42 km/h and from town B to town A at a speed of 48 km/h. What is the average speed of the car?

    a) 45 km/h b) 46 km/h c) 44 km/h
    d) Cannot be determined e) None of the above

45. A man divides his total route of journey into three equal parts and decides to travel the three parts with speed of 15 km/h, 10 km/h and 5 km/h, respectively. Find his average speed of during journey.

    a) 2.28 km/h b) 9 km/h c) 14 km/h d) 8.18 km/h e) None of the above

46. Mr. Bundda travels in his car and covers 1/4 part of his journey with 8 km/h, 3/5 part with 6 km/h and remaining 3/20 part with a speed of 10 km/h. Find out his average speed during the whole journey.

    a) 6.83 km/h b) 9 km/h c) 4 km/h d) 8.5 km/h e) None of the above

47. A man spends an average of ₹1694. 70 per month for the first 7 month and ₹1810.50 per month for the next 5 months. His monthly salary if he saves ₹3084.60 during the whole year is

    a) ₹2400 b) ₹3000 c) ₹1000 d) ₹2000 e) None of the above

48. The average marks of 120 students are 35. If the average of passed students was 39 and failed students was 15, then find the number of students who have passed.

    a) 80 b) 100 c) 120 d) 140 e) None of the above

49. A person covers 9 km with a speed of 3 km/h, 25 km with a speed of 5 km/h and 30 km with a speed of 10 km/h. Find out the average speed of person.

    a) $5\frac{9}{11}$ km/h b) $11\frac{5}{9}$ km/h c) $9\frac{5}{11}$ km/h d) $5\frac{5}{11}$ km/h e) None of the above

50. In the afternoon, a student read 100 pages at the rate of 60 pages/h. In the evening, when she was tired, she read 100 more pages at the rate of 40 pages/h. What was her average rate of reading, in pages per hour?

    a) 48 b) 50 c) 60 d) 70

51. Five years ago, the average age of P and Q was 15 yr. Now, average age of P, Q and R is 20 yr. What would be the age of R after 10 yr?

    a) 35 yr b) 40 yr c) 30 yr d) 50 yr

52. The average of four positive integers is 73.5. The highest integer is 108 and the lowest integer is 29. The difference between the remaining two integers is 15. Which of the following is the smaller of the remaining two integers?

    a) 80 b) 86 c) 73 d) Cannot be determined e) None of the above

53. The daily minimum temperature of a place in a certain week, beginning from Monday was 20°, 18.6°, 17°, 21.4°, 19° and 19.8°. It was known that the average of the whole week was 19.5°. The minimum temperature on Sunday was

    a) 16.9° b) 11.6° c) 20.7° d) 12.8°

54. The average age of two boys and their father is greater than the average age of those two boys and their mother by 3 yr. The average age of the four is 19 yr, If the average age of the two boys be $5\frac{1}{2}$yr, then find the age of the father and mother.

    a) 37 yr and 28 yr b) 47 yr and 38 yr c) 50 yr and 41 yr

    d) 35 yr and 32 yr e) None of the above

55. The average salary of the entire staff in an office is ₹500 per day. The average salary of officers is rupee 750 and that of non-officers is ₹250. If the number of officers is 15, then find the number of non-officers in the office.

    a) 25 b) 10 c) 15 d) 20

Exercise : Higher Skill Level Questions

1. A has double the money of B and B has 50% more money than C. If average money of all the three persons is ₹12000, how much money does A have?

    a) ₹$\frac{211000}{11}$ b) ₹$\frac{315000}{11}$ c) ₹$\frac{216000}{11}$ d) ₹$\frac{316000}{11}$ e) None of the above

2. A cricketer scored some runs in his continuous 9 innings. He scored 100 runs in his 10th innings and increases his average by 8 runs. What was the average of his runs at the end of 10th innings?

    a) 20 b) 24 c) 28 d) 32 e) None of the above

3. Three year ago, the average age of a family of five members was 27 yr. The present average age is also 27 yr after the inclusion of a child into the family. Find the age of the child.

   a) 16 yr b) 12 yr c) 24 yr d) 20 yr

4. Average weight of three boys P, T and R is $54\frac{1}{3}$ kg, while the average weight of three boys T,F and G is 53 kg. What is the average weight of P,T,R,F and G

   a) 53.8 kg b) 52.4 kg c) 53.2 kg d) Cannot be determined e) None of the above

5. The average runs scored by a cricketer in 42 innings, is 30. The difference between his maximum and minimum scores in an innings is 100. If these two innings are not taken into consideration, then the average score of remaining 40 innings is 28. Calculate the maximum runs scored by him in an innings?

   a) 125 b) 120 c) 110 d) 100

6. The average age of 5 sisters is 20 yr. If the age of the youngest sister be 4 yr, what was the average age of the group of sisters at the birth of the youngest sister?

   a) 25 yr b) 15 yr c) 18 yr d) 20 yr e) None of the above

7. The average marks obtained by 120 students in an examination is 30. If the average marks of passed students is 40 and that of the failed students is 10, what is the 25% of the number of students who passed the examination?

   a) 30 b) 20 c) 40 d) 15 e) None of the above

8. A cricketer played 80 innings and scored an average of 99 runs. His score in the last inning is zero run. To have an average of 100 at the end, his score in the last innings should have been

   a) 10 runs b) 1 run c) 60 runs d) 80 runs

9. If $x + y + z - 13$, then the maximum value of $(x-2)(y+1)(z-3)$ is

   a) 25 b) 30 c) 54 d) 27

10. A class is divided into two sections A and B. Passing average of 20 students of section A is 80% and passing average of 30 students of section B is 70%. What is the passing average of both the sections?

    a) 72% b) 74% c) 75% d) 77%

11. The average age of 4 members of a family is 20 yr. If youngest member is 4 yr old, what was the average age of family at the time of the birth of youngest member?

    a) 27 yr b) $23\frac{1}{3}$ yr c) $21\frac{1}{3}$ yr d) 22 yr e) None of the above

12. The arithmetic mean of the scores of a group of students in a test was 52. The brightest 20% of them secured a mean score of 80 and the dullest 25%, a mean score of 31. The mean score of remaining 55% is

a) 54.6% b) 45% c) 50% d) 51.4% e) None of the above

13. A car reached Raipur from Somgarh in 35 min with an average speed of 69 km/h. If the average speed is increased by 36 km/h, how much time will it take to cover the same distance?

    a) 24 min b) 27 min c) 23 min d) 29 min e) None of the above

14. The average weight of 4 men A, B, C and D, is 67 kg. The 5th man E is included and the average weight decreases by 2 kg. A is replaced by F. The weight of F is 4 kg more than E. Average weight decreases because of the replacement of A and now the average weight is 64 kg. Find the weight of A.

    a) 78 kg b) 66 kg c) 75 kg d) 58 kg e) None of the above

15. 3 yr ago, the average age of a family of 5 members was 17 yr. Even after the birth of a child, the average is same today. Find out the age of child.

    a) 1 yr b) 3 yr c) $2\frac{1}{2}$ yr d) 2 yr e) None of the above

16. The average age of three boys is 15 yr. If the ratio of their ages is 3 : 5 : 7, what is the age of the oldest boy?

    a) 7 yr b) 14 yr c) 20 yr d) 21 yr e) None of the above

17. 5 yr ago, the average age of Rashmi and Surbhi was 20 yr. Now, the average age of Rashmi, Surbhi and Meeta is 30 yr. What will be the age of Meeta after 15 yr?

    a) 35 yr b) 55 yr c) 59 yr d) 67 yr e) None of the above

18. The average weight of 60 students in a class was calculated to be 40 kg. Later, it was found out that the weight of one of the students was calculated as 36 kg, whereas his actual weight was 33 kg. What is the actual average weight of the students in the class?

    a) 39.90 kg b) 39.95 kg c) 39 kg d) 38 kg e) None of the above

19. A person bought some oranges worth ₹36 from each of the five markets at ₹1, ₹1.50, ₹1.80, ₹2 and ₹2.25 per orange, respectively. What is the average price of an orange?

    a) ₹1.91 b) ₹2.00 c) ₹1.58 d) ₹1.80 e) None of the above

20. Average of a, b and c is 11; average of c, d and e is 17; average of e and f is 22 and average of e and c is 17. Find out the average of a, b, c, d, e and f.

    a) $15\frac{2}{3}$ b) $18\frac{1}{2}$ c) $21\frac{1}{3}$ d) $16\frac{1}{2}$

21. The average age of 54 girls in a class was calculated as 14 yr. It was later realized that the actual age of one of the girls in the class was 10.5 yr but it was calculated as 13 yr. What is the actual average age (round off) of the girls in the class? (Find the approximate answer.)

    a) 10.5 yr b) 12.5 yr c) 12 yr d) 14 yr e) None of the above

22. The average age of a man and his two twin sons born on the same day is 30 yr. The ratio of the ages of father and one son is 5 : 2, what is the father's age?

    a) 50 yr b) 30 yr c) 45 yr d) 20 yr e) None of the above

23. The ratio of pens and pencils in a stationary shop is 3 : 2, respectively. The average number of pens and pencils is 180. What is the 30% of number of pencils in the shop?

    a) 150 b) 144 c) 245 d) 115 e) None of the above

24. If the average of the ages of Rakesh and Mohan is 15, average of the ages of Mohan and Ramesh is 12 and the average of the ages of Rakesh and Ramesh is 13, then the age of Mohan is

    a) 16 b) 13 c) 14 d) 12 e) None of the above

25. A cricketer has a certain average for 10 innings. In the eleventh innings, he scored 216 runs, there by increasing his average by 12 runs. Find out his new average.

    a) 96 b) 84 c) 97 d) 87 e) None of the above

26. A cricketer whose bowling average is 12.4 runs per wicket, takes 5 wickets for 26 runs and there by decreases his average by 0.4. Find out the number of wickets taken by him till the last match.

    a) 75 b) 81 c) 85 d) 79 e) None of the above

27. Nine friends have a dinner in a hotel. Eight of them spent ₹12 each on their meals and the ninth spent ₹16 more than the average expenditure of all the nine. Find out the total money spent by them.

    a) ₹126 b) ₹135 c) ₹111 d) ₹141 e) None of the above

28. The average weight of the students in four sections A, B, C and D is 60 kg. The average weight of the students of A, B, C and D individually are 45 kg, 50 kg, 72 kg and 80 kg, respectively. If the average weight of the students of section A and B together is 48 kg and that of B and C together is 60 kg, what is the ratio of the number of students in sections A and D?

    a) 12 : 7 b) 4 : 3 c) 3 : 2 d) 8 : 5 e) None of the above

29. The average age of students of a class is 15.8 yr. The average age of boys in the class is 16.4 yr and that of the girls is 15.4 yr. Find out the ratio of the number of boys to the number of girls in the class.

    a) 3 : 1 b) 5 : 2 c) 2 : 3 d) 3 : 7

30. In a certain examination, the average marks of an examinee is 64 per paper. If he had obtained 18 more marks for his Mathematics paper and 4 more marks for his English paper, his average per paper would have been 66. How many papers were there in the examination?

    a) 11 b) 13 c) 9 d) 10 e) None of the above

Basic Skill Level Questions Answer Key																				
1	e	6	c	11	c	16	c	21	d	26	b	31	a	36	a	41	d	46	a	51	c
2	e	7	a	12	b	17	d	22	b	27	a	32	d	37	c	42	d	47	d	52	e
3	d	8	e	13	a	18	b	23	a	28	b	33	e	38	a	43	b	48	b	53	c
4	b	9	b	14	a	19	b	24	a	29	b	34	d	39	c	44	e	49	a	54	a
5	c	10	d	15	d	20	a	25	c	30	d	35	b	40	a	45	d	50	a	55	c

Higher Skill Level Questions											
1	c	6	d	11	c	16	d	21	d	26	c
2	c	7	b	12	d	17	b	22	a	27	a
3	b	8	d	13	c	18	b	23	e	28	b
4	d	9	d	14	b	19	c	24	c	29	c
5	b	10	b	15	d	20	a	25	a	30	a

Chapter 8- Ratio and Proportion

1. If $P : Q = 8 : 15$ and $Q : R = 3 : 2$, then find $P : Q : R$.
   a) 8 : 15 : 7 b) 7 : 15 : 8 c) 8 : 15 : 10 d) 10 : 15 : 8 e) None of the above
2. If $P : Q = 8 : 15$, $Q : R = 5 : 8$ and $R : S = 4 : 5$, then $P : S$ is equal to
   a) 4 : 15 b) 2 : 15 c) 3 : 19 d) 7 : 15 e) None of the above
3. If $A : B = 2 : 3$, $B : C = 5 : 7$ and $C : D = 3 : 10$, then what is $A : D$ equal to?
   a) 1 : 7 b) 2 : 7 c) 1 : 5 d) 5 : 1
4. Find the 4th proportional to 4, 16 and 7.
   a) 28 b) 29 c) 22 d) 25 e) None of the above
5. Calculate the 3rd proportional to 15 and 30.
   a) 55 b) 15 c) 65 d) 60 e) None of the above
6. Find the mean proportional between 9 and 64.
   a) 25 b) 24 c) 27 d) 35 e) None of the above
7. What is the mean proportional between $(15 + \sqrt{200})$ and $(27 - \sqrt{648})$?
   a) 4 b) $14\sqrt{7}$ c) $3\sqrt{5}$ d) $5\sqrt{3}$
8. What will be the duplicate ratio of 2 : 7?
   a) 4 : 49 b) 49 : 4 c) 4 : 14 d) 8 : 343 e) None of the above
9. Find the sub-duplicate ratio of 81 : 64.
   a) 8 : 9 b) 4 : 9 c) 9 : 8 d) 7 : 8 e) None of the above
10. Find the triplicate ratio of 7 : 5.
    a) 125 : 343 b) 343 : 125 c) 344 : 125 d) 343 : 126 e) None of the above
11. Calculate the sub-triplicate ratio of 512 : 729.
    a) 9 : 8 b) 4 : 9 c) 7 : 8 d) 8 : 9 e) None of the above
12. What will be the inverse ratio of 17 : 19?
    a) 19 : 17 b) 18 : 17 c) 17 : 18 d) 19 : 5 e) None of the above
13. Find the compound ratio of 2 : 7, 5 : 3 and 4 : 7.
    a) 147 : 40 b) 40 : 147 c) 147 : 30 d) 30 : 147 e) None of the above

14. If $A : B = 3 : 4$ and $B : C = 8 : 9$, then find the value of $A : B : C$.

a) 3 : 4 : 5 b) 1 : 2 : 3 c) 7 : 12 : 17 d) 6 : 8 : 9 e) None of the above

15. If $a : b = b : c$, then ratio $a^4 : b^4$ is equal to

a) $ac : b^2$ b) $a^2 : c^2$ c) $c^2 : a^2$ d) $b^2 : ac$ e) None of the above

16. If $a : b = 3 : 5$ and $b : c = 4 : 7$, then $a : c$ is equal to

a) 11 : 35 b) 35 : 11 c) 35 : 12 d) 12 : 35 e) None of the above

17. If $4a = 5b$ and $7b = 9c$, then $a : b : c$ is equal to

a) 45 : 36 : 28 b) 44 : 33 : 28 c) 28 : 36 : 45

d) 36 : 28 : 45 e) None of the above

18. If $\frac{a}{7} : \frac{b}{9} = \frac{c}{11}$, then find $a : b : c$

a) 11 : 9 : 7 b) 9 : 7 : 11 c) 7 : 9 : 11 d) 11 : 7 : 9 e) None of the above

19. If 30% of A = 20% of B, then find the value of $A : B$.

a) 1 : 3 b) 3 : 2 c) 3 : 1 d) 2 : 3 e) None of the above

20. If $P : Q : R = 2 : 3 : 4$, then find $\frac{P}{Q} : \frac{Q}{R} : \frac{R}{P}$

a) 8 : 9 : 24 b) 9 : 8 : 24 c) 24 : 8 : 9 d) 8 : 24 : 9 e) None of the above

21. If $A = \frac{1}{4}B$ and $B = \frac{1}{2}C$, then find the value of $A : B : C$

a) 8 : 4 : 1 b) 4 : 2 : 1 c) 1 : 4 : 8 d) 1 : 2 : 4

22. If $0.8 \times A = 0.09 \times B$, then find $A : B$.

a) 9 : 80 b) 3 : 80 c) 80 : 9 d) 7 : 80 e) None of the above

23. $\frac{1}{x} : \frac{1}{y} : \frac{1}{z} = 2 : 3 : 5$ then determine $x : y : z$

a) 6 : 15 : 10 b) 3 : 15 : 10 c) 15 : 3 : 10 d) 15 : 10 : 6 e) None of the above

24. If $\frac{a}{3} = \frac{b}{8}$, then $(a + 3) : (b + 8)$ is equal to

a) 3 : 8 b) 8 : 3 c) 5 : 8 d) 3 : 5 e) None of the above

25. If $(a + b) : (a - b) = 5 : 3$, then find $(a^2 + b^2) : (a^2 - b^2)$.

a) 17 : 5 b) 25 : 9 c) 4 : 1 d) 16 : 1 e) None of the above

26. If $\frac{x}{2y} = \frac{6}{7}$, then find the value of $\frac{x-y}{x+y} + \frac{14}{19}$

a) 5 b) 1 c) 4 d) 3 e) None of the above

27. If $\frac{1}{2}$ of $A = \frac{2}{5}$ of $B = \frac{1}{3}$ of C, then $A : B : C$ is equal to

a) 4 : 5 : 6 b) 6 : 4 : 5 c) 5 : 4 : 6 d) 4 : 6 : 5 e) None of the above

28. If $P^2 + 4Q^2 = 4PQ$, then determine $P : Q$.

a) 1 : 3 b) 3 : 1 c) 2 : 1 d) 3 : 2 e) None of the above

29. If $xy = 36$, then which of the following is correct?

a) $x : 9 = 4 : y$ b) $9 : x = 4 : y$ c) $x : 17 = y : 7$

d) $x : 6 = y : 6$ e) None of the above

30. If 10% of $(A + B)$ = 50% of $(A - B)$, then find $A : B$.

a) 1 : 2 b) 5 : 2 c) 2 : 3 d) 3 : 2 e) None of the above

31. If $a : b = c : d = e : f = 1 : 2$, then find $(3a + 5c + 7e) : (3b + 5d + 7f)$.

a) 1 : 2 b) 2 : 1 c) 3 : 1 d) 1 : 3 e) None of the above

32. If $x : y = 7 : 5$, then what is the value of $(5x - 2y) : (3x + 2y)$

a) $\frac{5}{4}$ b) $\frac{6}{5}$ c) $\frac{25}{31}$ d) $\frac{31}{42}$

33. If a, b, c, d and e are in continued proportion, then $a : e$ is equal to

a) $\frac{a^3}{b^3}$ b) $\frac{a^4}{b^4}$ c) $\frac{b^3}{a^3}$ d) $\frac{b^4}{a^4}$

34. If $2A = 3B = 4C$, then find $A : B : C$.

a) 2 : 3 : 4 b) 4 : 3 : 2 c) 6 : 4 : 3 d) 3 : 4 : 6 e) None of the above

35. In a certain school, the ratio of boys to girls is 7 : 5. If there are 2400 students in the school, then how many girls are there in the school?

a) 500 b) 700 c) 800 d) 1000

36. If $a + b : b + c : c + a = 6 : 7 : 8$ and $a + b + c = 14$, then find c.

a) 6 b) 7 c) 8 d) 10 e) 14

37. The quantity that must be added to each term of $a : b$, so as to make it $c : d$, is

a) $\frac{ab - cd}{a - b}$ b) $\frac{ac + bd}{c + a}$ c) $\frac{ad + bc}{c + d}$ d) $\frac{ad - bc}{c - d}$

38. One-half of a certain number is equal to 65% of the 2nd number. Find the ratio of 1st to 2nd number.

    a) 10 : 13 b) 8 : 13 c) 13 : 8 d) 13 : 10 e) None of the above

39. The ratio between two numbers is 3 : 4. If each number is increased by 3, the ratio becomes 4 : 5. Find the difference between the numbers.

    a) 3 b) 9 c) 2 d) 7 e) None of the above

40. From each of two given numbers, half the smaller number is subtracted. After such subtraction, the larger number is 4 times as large as the smaller number. What is the ratio of the numbers?

    a) 5 : 2 b) 1 : 4 c) 4 : 1 d) 4 : 5

41. Two numbers are in the ratio of 4 : 7. If 5 is subtracted from each, the ratio becomes 1 : 2. Find the greater number.

    a) 15 b) 40 c) 20 d) 35

42. Two numbers are in the ratio 2 : 3. If 9 is added to each number, they will be in the ratio 3 : 4. What is the product of the two numbers?

    a) 360 b) 480 c) 486 d) 512 e) None of the above

43. The total number of students in a school is 2140. If the number of girls in the school is 1200, then what is the ratio of the total number of boys to the total number of girls in the school?

    a) 26 : 25 b) 47 : 60 c) 18 : 13 d) 31 : 79 e) None of the above

44. The total weight of Sanjay and Suresh is 120 kg. If Sanjay weighs 30 kg more than Suresh, then what is the ratio of the weight of Suresh to that of Sanjay?

    a) 0.4 b) 0.6 c) 0.3 d) 0.25 e) None of the above

45. In a school, the ratio of boys and girls is 4 : 5. When 100 girls leave the school, the ratio becomes 6 : 7. How many boys are there in the school?

    a) 1800 b) 1200 c) 1000 d) 1500 e) None of the above

46. In a class, the number of boys and girls is in the ratio of 4 : 5. If 10 more boys join the class, the ratio of numbers of boys and girls becomes 6 : 5. How many girls are there in the class?

    a) 20 b) 30 c) 25 d) Couldn't be determined e) None of the above

47. The total number of students in a school is 8670. If the number of boys in the school is 4545, then what will be the ratio of the total number of boys to the total number of girls in the school?

    a) 303 : 275 b) 275 : 303 c) 11 : 12 d) 12 : 11 e) None of the above

48. A truck covers a distance of 640 km in 10 h. A car covers the same distance in 8 h. What is the respective ratio of the speed of the truck to that of car?

    a) 3 : 4 b) 1 : 2 c) 5 : 6 d) 6 : 7 e) None of the above

49. A certain distance is covered at a certain speed . If half of this distance is covered in double the time, the ratio of the two speeds is

    a) 4 : 1 b) 1 : 4 c) 2 : 1 d) 1 : 2

50. The speeds of three cars are in the ratio of 2:3:4 Find the ratio between the time taken by these cars to cover the same distance.

    a) 2 : 3 : 4 b) 4 : 3 : 2 c) 4 : 3 : 6 d) 6 : 4 : 3 e) None of the above

51. A person distributes his pens among four friends A, B, C and D in the ratio of $\frac{1}{3} : \frac{1}{4} : \frac{1}{5} : \frac{1}{6}$. What is the minimum number of pens that the person should have?

    a) 75 b) 45 c) 57 d) 65

52. Divide ₹990 into 3 parts in such a way that half of the first part, one-third of the second part and one-fifth of the third part are equal

    a) 198, 494, 298 b) 198, 297, 495 c) 200, 300, 490 d) 196, 298, 496

53. A bag contains ₹1, 50 paise and 25 paise coins in the ratio of 8 : 9 : 11. If the total money in the bag is ₹366, then find the number of 25 paise coins.

    a) 264 b) 364 c) 241 d) 245 e) None of the above

54. Weekly incomes of two persons are in the ratio of 7 : 3 and their weekly expenses are in the ratio of 5 : 2. If each of them saves ₹300 per week, then the weekly income of the first person is

    a) ₹7500 b) ₹4500 c) ₹6300 d) ₹5400 e) None of the above

55. Amit and Sudesh have invested in the ratio of 4 : 7. If both invested a total amount of ₹49500, then find the investment of Sudesh.

    a) ₹31500 b) ₹18000 c) ₹31000 d) ₹18500 e) None of the above

56. If a sum of ₹1664 is divided between P and Q in the ratio of $1/3 : 1/5$, then find P's share.

    a) 1085 b) 1015 c) 1090 d) 1040 e) None of the above

57. The marks of 3 students A, B and C are in the ratio 10 : 12 : 15. If the maximum marks of the paper are 100, then the marks of B cannot be in the range of

    a) 20-30 b) 40-50 c) 70-80 d) 80-90 e) None of the above

58. A sum of ₹7000 is divided among A, B and C in such a way that the shares of A and B are in the ratio 2 : 3 and those of B and C are in the ratio 4 : 5. The share of B is

    a) ₹1600 b) ₹2000 c) ₹2400 d) ₹3000 e) None of the above

59. A sum of ₹300 is divided among P, Q and R in such a way that Q gets ₹30 more than P and R gets ₹60 more than Q. Then, ratio of their shares is

    a) 2 : 3 : 5 b) 3 : 2 : 5 c) 2 : 5 : 3 d) 5 : 3 : 2

60. A sum of money is divided amongst A, B, C and D in the ratio of 3 : 7 : 9 : 13. If the share of B is ₹4872, then what will be the total amount of money of A and C together?

a) ₹8352 b) ₹6998 c) ₹9784 d) ₹7456 e) None of the above

61. What sum of money is to be divided among 3 persons in the ratio 3 : 4 : 7, so that the second person receives ₹12 only?

a) ₹21 b) ₹32 c) ₹36 d) ₹42

62. The prices of a scooter and a television are in the ratio of 8 : 7. If a scooter costs ₹2000 more than a television, then find out the price of television.

a) ₹14000 b) ₹28000 c) ₹10000 d) ₹18000 e) None of the above

63. The ratio between the ages of A and B is 3: 5 and the sum of their ages is 56 yr. The ratio between their ages 7 yr ago was

a) 3 : 4 b) 3 : 5 c) 1 : 2 d) 2 : 3

64. The ratio of the ages of a father to that of his son is 5 : 2. If the product of their ages (in years) is 1000, then find the father's age after 10 yr.

a) 50 yr b) 60 yr c) 80 yr d) 100 yr

65. A certain number is divided into two parts such that 5 times the first part added to 11 times the second part makes 7 times the whole. The ratio of the first part to the second part is

a) 2 : 1 b) 5 : 11 c) 1 : 2 d) 2 : 3

66. 35% of a number is two times 75% of another number. What is the ratio between the first and the second number, respectively?

a) 35 : 6 b) 31 : 7 c) 23 : 7 d) 32 : 9 e) None of the above

67. Brothers A and B had some savings in the ratio 4 : 5. They decided to buy a gift for their sister, sharing the cost in the ratio 3 : 4. After they bought, A spent two-third of his amount, while B is left with ₹145. Then, the value of the gift is

a) ₹70 b) ₹105 c) ₹140 d) ₹175

Exercise : Higher Skill Level Questions

1. 1. In a town, 80% of the population are adults of which the men and women are in the ratio of 9 : 7, respectively. If the number of adult women is 4.2 lakh, what is the total population of the town?

a) 12 lakh b) 9.6 lakh c) 9.8 lakh d) 11.6 lakh e) None of the above

2. Mr. Shrimant inherits 2505 gold coins and divides them among his three sons; Bharat, Parat and Marat; in a certain ratio. Out of the total coins received by each of them, Bharat sells 30 coins, Parat donates his 30 coins and Marat losses 25 coins. Now, the ratio of gold coins with them is 46 : 41 : 34, respectively. How many coins did Parat receive from his father?

   a) 705 b) 950 c) 800 d) 850 e) None of the above

3. ₹600 are divided among A, B and C such that ₹40 more than 2/5th of A's share, ₹20 more than 2/7th of B's share and ₹10 more than 9/17th of C's share, all are equal. What is B's share?

   a) ₹150 b) ₹185 c) ₹280 d) ₹285 e) None of the above

4. Salaries of Akash, Bablu and Chintu are in the ratio of 2 : 3 : 5. If their salaries were increased by 15%, 10% and 20% respectively, then what will be the new ratio of their salaries?

   a) 3 : 3 : 10 b) 23 : 33 : 60 c) 20 : 22 : 40 d) None of the above

5. Salary of Mr. X is 80% of the salary of Mr. Y and the salary of Mr. Z is 120% of the salary of Mr. X. What is the ratio between the salaries of X, Y and Z, respectively?

   a) 4 : 6 : 5 b) 4 : 5 : 6 c) 16 : 24 : 25 d) 16 : 25 : 24 e) None of the above

6. If the positions of the digits of a two-digit number are interchanged, the number newly formed is smaller than the original number by 45. Also, the ratio of the new number to the original number is 3 : 8. What is the original number?

   a) 61 b) 72 c) 94 d) Couldn't be determined e) None of the above

7. A cat takes 5 leaps for every 4 leaps of a dog but 3 leaps of the dog are equal to 4 leaps of the cat. What is the ratio of the speeds of the cat to that of the dog?

   a) 11 : 15 b) 15 : 11 c) 16 : 15 d) 15 : 16

8. In the month of January, Aran's income and expenses were ₹15000 and ₹9000, respectively. His monthly expenses vary directly as the square of his monthly income. What is his income when it just equals his expenses?

   a) ₹25000 b) ₹20000 c) ₹35000 d) ₹15000 e) None of the above

9. The respective ratio of Sita's, Riya's and Kunal's monthly incomes is 84 : 76 : 89. If Riya's annual income is ₹456000, then what is the sum of Sita's and Kunal's annual incomes? (In some cases monthly income and in some cases annual income is used.)

   a) ₹195000 b) ₹983500 c) ₹1130000 d) ₹1038000 e) None of the above

10. ₹5625 are divided among A, B and C, so that A receives 1/2 as much as B and C together receive and B receives 1/4 as much as A and C together receive. Find the sum of shares of A and B.

    a) ₹5000 b) ₹3000 c) ₹15000 d) ₹9000 e) None of the above

11. ₹710 were divided among A, B and C in such a way that A had ₹40 more than B and C had ₹30 more than A. How much was C's share?

    a) ₹270 b) ₹300 c) ₹135 d) ₹235 e) None of the above

12. The ratio of 1st and 2nd classes train fairs between two stations is 3 : 1 and that of the number of passengers travelling between these stations by 1st and 2nd classes is 1 : 50. If on a particular day, ₹2650 be collected from the passengers travelling between these stations, then find the amount collected from 2nd class passengers.

    a) ₹3000 b) ₹3500 c) ₹2800 d) ₹2500 e) None of the above

13. Out of two sections A and B, 10 students of section B shift to A, as a result strength of A becomes 3 times the strength of B. But, if 10 students shift over from A to B, both A and B become equal in strength. Ratio of the number of students in section A that of section B is [SSC CPO 2013]

    a) 2 : 1 b) 5 : 3 c) 3 : 1 d) 9 : 4

14. A sum of money is to be divided equally among P, Q and R in the respective ratio of 5 : 6 : 7 and another sum of money is to be divided between S and T equally. If S got ₹2100 less than P, then how much amount did Q receive?

    a) ₹2500 b) ₹2000 c) ₹1500 d) Couldn't be determined e) None of the above

15. Nandita scores 80% marks in five subjects together, viz., Hindi, Science, Mathematics, English and Sanskrit, where in the maximum marks of each subject were 105. How many marks did Nandita score in Science, if she scored 89 marks in Hindi, 92 marks in Sanskrit, 98 marks in Mathematics and 81 marks in English?

    a) 60 b) 75 c) 65 d) 70 e) None of the above

16. ₹2186 are distributed among A,B and C. If money given to them is decreased by ₹26, ₹28 and ₹32 respectively, then they have money in the ratio of 9 : 13 : 8. What is the amount given to A?

    a) ₹575 b) ₹640 c) ₹656 d) ₹672 e) None of the above

17. A person gave 2/5 part of his income to his elder son and 30% part to his younger son. He saved his remaining money in three trusts A, B and C in the ratio of 3 : 5 : 2. If difference between the amount got by his both sons is ₹2000, how much amount he saved in trust C?

    a) ₹1000 b) ₹1140 c) ₹1200 d) ₹1256 e) ₹1300

18. 18. In a factory, the ratio of the numbers of employees of three types A, B and C is 9 : 13 : 18 and their wages are in the ratio of 10 : 7 : 4. If number of employees of type C is 54 and wages of every employee of type B is ₹1400, then find the total wages of all the employees of type A.

    a) ₹51000 b) ₹54000 c) ₹56000 d) ₹57000 e) ₹59000

19. Number of employees in a factory decreases in the ratio of 8 : 7 and salary of employees increases in the ratio of 5 : 6. Find wheather the total salary given to the empolyees is increased or decreased and in what ratio?

a) 19 : 20, increased b) 20 : 19, decreased c) 19 : 21, increased

d) 21 : 20, decreased e) None of the above

20. Out of 120 applications for a post, 70 are males and 80 have a driver's license. What is the ratio between the minimum to maximum number of males having driver's license ?

a) 1 : 2 b) 2 : 3 c) 3 : 7 d) 5 : 7 e) None of the above

21. In a certain examination, the number of those who passed was 4 times the number of those who failed. If there had been 35 fewer candidates and 9 more had failed, the ratio of passed and failed candidates would have been 2 : 1, then the total number of candidates was

a) 135 b) 155 c) 145 d) 150

Basic Skill Level Questions Answer Key

1	c	11	d	21	c	31	a	41	d	51	c	61	d
2	a	12	a	22	a	32	c	42	c	52	b	62	a
3	a	13	b	23	d	33	b	43	b	53	a	63	c
4	a	14	d	24	a	34	c	44	b	54	c	64	b
5	d	15	b	25	a	35	d	45	b	55	a	65	a
6	b	16	d	26	b	36	a	46	c	56	d	66	e
7	c	17	a	27	a	37	d	47	a	57	d	67	c
8	a	18	c	28	c	38	d	48	e	58	c		
9	c	19	d	29	a	39	a	49	b	59	a		
10	b	20	a	30	d	40	a	50	d	60	a		

Higher Skill Level Questions

1	a	6	b	11	a	16	c	21	b
2	d	7	d	12	d	17	c		
3	c	8	a	13	a	18	b		
4	b	9	d	14	d	19	e		
5	e	10	b	15	a	20	c		

Chapter 9- Problem Based on Ages

1. The present ages of Reena and Usha are 24 and 36 yr, respectively. What was the ratio between the ages of Usha and Reena respectively 8 yr ago?

   a) 7 : 4 b) 4 : 7 c) 11 : 8 d) 8 : 11 e) None of the above

2. Raju decided to marry 3 yr after he gets a job. He was 17 yr old when he passed class 12th. After passing class 12th, he had completed his graduation course in 3 yr and PG course in 2 yr. He got the job exactly 1 yr after completing his PG course. At what age will he get married?

   a) 27 yr b) 26 yr c) 28 yr d) 23 yr e) None of the above

3. The average age of 8 children of a family is 12 yr. If the age of 7 children are 12, 8, 14, 11, 9, 13 and 15 yr. Then the age of 8th child will be

   a) 12 yr b) 14 yr c) 13 yr d) 15 yr

4. The ratio between the present ages of A and B is 4 : 5. If the ratio between their ages 4 yr hence becomes 14 : 17, then what is B's age at present?

   a) 30 yr b) 28 yr c) 34 yr d) Data inadequate

5. In a family, the average age of father and mother is 35 yr. The average age of the father, mother and their only son is 27 yr. What is the age of the son?

   a) 12 yr b) 11 yr c) 10.5 yr d) 10 yr

6. The sum of the present ages of Varun and Kapil is 42 yr. The ratio of their ages after 5 yr will be 15 : 11. What is the present age of Kapil?

   a) 17 yr b) 24 yr c) 25 yr d) 22 yr

7. The average age of 9 students and their teacher is 16 yr. The average age of the first four students is 19 yr and that of the last five is 10 yr. The teacher's age is

   a) 36 yr b) 34 yr c) 30 yr d) 28 yr

8. At present, Anil is 1.5 times of Purvi's age. 8 yr hence, the respective ratio between Anil and Purvi's ages will be 25 : 18 What is Purvi's present age?

   a) 50 yr b) 28 yr c) 42 yr d) 36 yr e) None of the above

9. The average of the present ages of Sachin and Saurabh is 36 yr. If Sachin is 8 yr older than Saurabh, what is the Saurabh's present age?

   a) 30 yr b) 34 yr c) 32 yr d) 40 yr e) None of the above

10. At present Meena is eight times her daughter's age. 8 yr from now, the ratio of the ages of Meena and her daughter will be 10:3, respectively. What is Meena's present age?

    a) 32 yr b) 40 yr c) 36 yr d) Cannot be determined e) None of the above

11. Raman's present age is three times his daughter's and nine-thirteenth of his mother's present age. The sum of the present ages of all three of them is 125 yr. What is the difference between the present ages of Raman's daughter and Raman's mother?

    a) 45 yr b) 40 yr c) 50 yr d) Cannot be determined e) None of the above

12. The ages of Harsh and Sumit are 40 yr and 60 yr, respectively. How many years before the ratio of their ages was 3 : 5?

    a) 15 yr b) 20 yr c) 37 yr d) 10 yr

13. The average age of a man and his son is 27 yr. The ratio of their ages is 8 : 1, respectively. What will be the son's age after 6 yr?

    a) 6 yr b) 14 yr c) 12 yr d) 8 yr e) None of the above

14. The difference between the ages of Meena and Seema is 3 yr and the ratio between their ages is 7 : 8. What is the sum of their ages?

    a) 43 yr b) 41 yr c) 45 yr d) 48 yr e) 47 yr

15. At present, the ratio of the ages of Maya and Chhaya is 6 : 5 and fifteen years from now, the ratio will get changed to 9 : 8. Maya's present age is

    a) 21 yr b) 24 yr c) 30 yr d) 40 yr

16. The ratio of the present age of Manoj to that of Wasim is 3 : 11. Wasim is 12 yr younger than Rehana. Rehana's age after 7 yr will be 85 yr. What is the present age of Manoj's father, who is 25 yr older than Manoj?

    a) 43 yr b) 67 yr c) 45 yr d) 69 yr e) None of the above

17. Before 7 yr, the ratio of ages of A and B was 3 : 4. After 9 yr, ratio of their ages will be 7 : 8. The present age of B will be

    a) 16 yr b) 19 yr c) 28 yr d) 23 yr e) None of the above

18. The ratio between the present ages of Indira and Lizzy is 3 : 8, respectively. After 8 yr, Indira's age will be 20 yr. What was Lizzy's age 5 yr ago?

    a) 37 yr b) 27 yr c) 28 yr d) 38 yr e) None of the above

19. The ratio between the present ages of Tarun and Varun is 3 : 7, respectively. After 4 yr, Varun's age will be 39 yr. What was Tarun's age 4 yr ago?

    a) 12 yr b) 13 yr c) 19 yr d) 18 yr e) None of the above

20. 7 yr ago, the ages of A and B were in the ratio 4 : 5 and 7 yr hence, they will be in the ratio 5 : 6. The present age of B is

    a) 56 yr b) 63 yr c) 70 yr d) 77 yr

21. At present, Kavita is twice Sarita's age. 8 yr hence, the respective ratio between Kavita's and Sarita's ages will be 22 : 13 What is Kavita's present age?

    a) 26 yr b) 18 yr c) 42 yr d) 36 yr e) None of the above

22. The average age of a family of five members is 24. If the present age of the youngest member is 8 yr, what was the average age of the family at the time of the birth of the youngest member?

    a) 20 yr b) 16 yr c) 12 yr d) 18 yr e) 21 yr

23. Four years ago, Shyam's age was 3/4 times of that of Ram. Four year hence, Shyam's age will be 5/6 times that of Ram. What is the present age of Shyam?

    a) 15 yr b) 20 yr c) 16 yr d) 24 yr e) 8 yr

24. The present ages of two persons are 36 and 50 yr, respectively. If after n yr the ratio of their ages will be 3 : 4, then the value of n is

    a) 3 b) 4 c) 7 d) 6

25. The ratio of the ages of Ram and Rahim 10 yr ago was 1 : 3 The ratio of their ages five years hence will be 2 : 3. Then, the ratio of their present ages is

    a) 1 : 2 b) 3 : 5 c) 3 : 4 d) 2 : 5

26. At present, Palash is three times Arnav's age. After 7 yr, Palash will be twice Arnav's age, then how many times will Palash's age be in another 14 yr time with respect to Arnav's age then?

    a) 1 b) 3 c) 2 d) 1.5 e) None of the above

27. 10 yr before, the ratio of ages of A and B was 13 : 17. After 17 yr from now, the ratio of their ages will be 10:11 The present age of B

    a) 23 yr b) 40 yr c) 27 yr d) 44 yr e) None of the above

28. 3 yr before, the average age of a family of 5 members was 17 yr. Due to birth of a new child, the average age is same today. The present age of child is

    a) 1 yr b) 3 yr c) $2\frac{1}{2}$ yr d) 2 yr

29. The age of a man after 15 yr is 4 times the age of that man 15 yr before. His present age is

    a) 10 yr b) 15 yr c) 20 yr d) 25 yr

30. The present age of Ravi's father is four times Ravi's present age. Five years back, Ravi's father was seven times as old as Ravi was at that time. What is the present age of Ravi's father?

    a) 84 yr b) 70 yr c) 40 yr d) 35 yr

31. Ravi's brother is 3 yr elder to him. His father was 28 yr of age when his sister was born, while his mother was 26 yr of age when he was born. If his sister was 4 yr of age when his brother was born, the ages of Ravi's father and mother respectively when his brother was born were

a) 32 yr and 23 yr
b) 32 yr and 29 yr
c) 35 yr and 29 yr
d) 35 yr and 33 yr

Basic Skill Level Questions Answer Key

1	a	6	a	11	c	16	a	21	d	26	a	31	a
2	b	7	b	12	d	17	d	22	a	27	c		
3	b	8	b	13	c	18	b	23	c	28	d		
4	a	9	c	14	c	19	e	24	d	29	d		
5	b	10	a	15	c	20	d	25	b	30	c		

Chapter 10 - Percentage

1. What is 5% of 50% of 500?

   a) 12.5 b) 25 c) 1.25 d) 6.25

2. If 20% of x = 90, find x.

   a) 350 b) 450 c) 250 d) 550 e) None of the above

3. 89% of ? + 365 = 1075.22

   a) 798 b) 897 c) 898 d) 752 e) None of the above

4. 50 kg is what per cent of 250 kg?

   a) 26% b) 15% c) 20% d) 18% e) None of the above

5. What is 32% of $\frac{3}{8}$th of 5000?

   a) 606 b) 610 c) 580 d) 600 e) None of the above

6. 12 of 26% of $\frac{5}{78}$ of 38% of $\frac{7}{152}$ of 10000=?

   a) 38 b) 35 c) 41 d) 52 e) None of the above

7. If 15% of A equal to 20% of B, then 25% of A is equal to what percent of B?

   a) 30% b) $33\frac{1}{3}$% c) 35% d) 25%

8. A number increased by $137\frac{1}{2}$% and the increment is 33. The number is

   a) 27 b) 22 c) 24 d) 25

9. A person's salary has increased from ₹7200 to ₹8100. What is the percentage increase in his salary

   a) 25% b) 18% c) $16\frac{2}{3}$% d) $12\frac{1}{2}$%

10. A's salary is 20% less than B's salary. Then B's salary is more than A's salary by

    a) $33\frac{1}{2}$% b) $16\frac{2}{3}$% c) 20% d) 25%

11. If the price of petrol is increased by 20%, by what percentage should the consumption be decreased by the consumer, if the expenditure on petrol remains unchanged?

    a) $16\frac{2}{3}$% b) $6\frac{2}{3}$% c) 8% d) 15% e) None of the above

12. If A's income is $\frac{5}{6}$th of B's income, then B's income is more than A's income by

    a) $\frac{25}{2}\%$ b) 15% c) 20% d) 25%

13. 25% of what amount of money is equal to $12\frac{1}{2}\%$ of ₹180

    a) ₹120 b) ₹75 c) ₹80 d) ₹90

14. 30% of a 3-digit number is 190.8. What will be 125% of that number?

    a) 759 b) 785 c) 795 d) 779 e) None of the above

15. The price of 10% increased of an article of ₹100, then again 10% increased the price. How much is increased in total price?

    a) 20 b) 21 c) 110 d) 121

16. In a company, there are 724 employees. 25% employees left the job. Find the number of rest of employees.

    a) 543 b) 550 c) 578 d) 565 e) None of the above

17. An agent sells goods of value of ₹15000. The commission which he receives at the ratio of $12\frac{1}{2}$ is

    a) ₹1875 b) ₹2125 c) ₹2000 d) ₹2700 e) None of the above

18. There are 1240 employees in an organisation. Out of which, 25% are promoted. How many such employees are there who got promotion?

    a) 398 b) 345 c) 310 d) 372 e) None of the above

19. The difference between 78% of a number and 59% of the same number is 323. What is 62% of that number?

    a) 1054 b) 1178 c) 1037 d) 1159 e) None of the above

20. The monthly income of a person is ₹5000. If his income is increased by 30%, then what is his monthly income now?

    a) ₹7000 b) ₹5500 c) ₹4500 d) ₹6500 e) None of the above

21. The price of a certain article is ₹15000. But due to slump in the market, its price decreases by 8%. Find the new price of the article.

    a) ₹14000 b) ₹13800 c) ₹16500 d) ₹12600 e) None of the above

22. A water pipe is cut into two pieces. The longer piece is 70% of the length of the pipe. By how much percentage is the longer piece longer than the shorter piece?

    a) 140% b) $\frac{400}{3}\%$ c) 40% d) None of the above

23. When 35 is subtracted from a number, it reduces to its 80%. What is $\frac{4}{5}$th of that number None of the above

    a) 70 b) 90 c) 120 d) 140 e) None of the above

24. A man losses 20% of his money. After spending 25% of the remainder, he has ₹480 left. What is the amount of money he originally had?

    a) ₹600 b) ₹720 c) ₹800 d) ₹840

25. If income of Vandana is 25% more than that of Aarti, then income of Aarti is how much per cent less than that of Vandana?

    a) 18% b) 21% c) 20% d) 24% e) None of the above

26. In an examination, the marks obtained by Shantanu is 40% less than the marks obtained by Kamal, then marks obtained by Kamal is how much per cent more than the marks obtained by Shantanu?

    a) $55\frac{2}{3}\%$ b) $44\frac{3}{5}\%$ c) $33\frac{1}{3}\%$ d) $66\frac{2}{3}\%$ e) None of the above

27. Because of scarcity of rainfall, the price of a land decreases by 12% and its production also decreases by 4%. What is the total effect on revenue?

    a) Loss of 16% b) Gain of 15% c) Loss of 15.48%
    d) Gain of 15.48% e) Loss of 15.52%

28. Mangulal, a shopkeeper, marks the prices of his goods at 20% higher than the original price. After that, he allows a discount of 5%. What profit or loss did he get?

    a) 14% (profit) b) 16% (loss) c) 14% (loss) d) 16% (profit) e) None of the above

29. The marked price of brand A watches is 15% higher than its original price. Due to increase in demand, the price is further increased by 10%. How much profit will be obtained in selling the watches?

    a) 25% b) 35% c) 26.5% d) 27% e) None of the above

30. The price of sugar is increased by 25%. If a family wants to keep its expenses on sugar unaltered, then the family will have to reduce the consumption of sugar by

    a) 20% b) 21% c) 22% d) 25%

31. If the price of tea falls down by 6%, by how much per cent must a householder increase its consumption, so as not to decrease expenditure?

    a) $5\frac{16}{47}\%$ b) $4\frac{18}{47}\%$ c) $6\frac{18}{47}\%$ d) $6\frac{17}{47}\%$ e) None of the above

32. A student has to score 40% marks to get through. If he gets 40 marks and fails by 40 marks, then find the maximum marks set for the examination.

    a) 200 b) 250 c) 300 d) 150 e) None of the above

33. A candidate scores 20% and fails by 50 marks, while another candidate who scores 40% marks, gets 30 marks more than the minimum required marks to pass the examination. Find the maximum marks for the examination.

    a) 500 b) 450 c) 300 d) 400 e) None of the above

34. The population of a town is 126800. It increase by 15% in the 1st year and decrease by 20% in the 2nd year. What is the population of the town at the end of 2 yr?

    a) 174984 b) 135996 c) 116656 d) 145820 e) None of the above

35. The population of a town is 705600. If it increases at the rate of 5% per annum, then what will be its population 2 yr hence?

    a) 777924 b) 777881 c) 778781 d) 797724 e) None of the above

36. The population of a town is 1058400. If it increases at the rate of 5% per annum, then find the population of the town 2 yr ago.

    a) 949000 b) 930000 c) 960000 d) 950000 e) None of the above

37. In an examination, 25% students failed in Mathematics, 20% failed in Economics and 5% failed in both. Find the percentage of students who passed in both the subjects.

    a) 60% b) 45% c) 39% d) 62% e) None of the above

38. The population of a city is 250000. It is increasing at the rate of 2% every year. The growth in the population after 2 yr is?

    a) 2500 b) 10000 c) 252000 d) 10100

39. In a school, 10% of boys are equal to the one-fourth of the girls. What is the ratio of boys and girls in that school?

    a) 3 : 2 b) 5 : 2 c) 2 : 1 d) 4 : 3

40. Income of Suman is first increased by 7% and then it is decreased by 7%. What is the change in her income?

    a) 0.49% (increase) b) 0.39% (decrease) c) 0.39% (increase)
    d) 0.49% (decrease) e) None of the above

41. 48% of the 1st number is 60% of the 2nd number. What is the ratio of the 1st number to the 2nd number?

    a) 4 : 7 b) 3 : 4 c) 5 : 4 d) Couldn't be determined e) None of the above

42. If the numerator of a fraction is increased by 200% and the denominator of the fraction is increased by 150%, the resultant fraction is $\frac{9}{35}$. What is the original fraction?

    a) $\frac{3}{10}$ b) $\frac{2}{15}$ c) $\frac{3}{14}$ d) $\frac{2}{7}$ e) None of the above

43. If length and breadth of a rectangle became half and double respectively, then what will be the resultant area?

    a) 25% b) 55% c) 75% d) 80% e) None of the above

44. Two numbers are respectively 20% and 50% more than a 3rd number. What is the percentage of 2nd with respect to 1st?

    a) 125% b) 90% c) 80% d) 75% e) None of the above

45. The sum of 15% of a positive number and 20% of the same number is 126. What is one-third of that number?

    a) 360 b) 1080 c) 120 d) 40 e) None of the above

46. In a test, A scored 10% more than B and B scored 5% more then C. If C scored 300 marks out of 400, then A's marks are

    a) 310 b) 325 c) 350 d) 360

47. In order to pass in an exam, a student is required to get 780 marks out of the aggregate marks. Sonu got 728 marks and was declared failed by 5%. What are the maximum aggregate marks a student can get in the examination?

    a) 1040 b) 1100 c) 1000 d) Cannot be determined e) None of the above

48. Mathew scored 42 marks in Biology, 51 marks in Chemistry, 58 marks in Mathematics, 35 marks in Physics and 48 marks in English. The maximum marks, a student can score in each subject, are 60. How much overall percentage did Mathew get in this exam?

    a) 76% b) 82% c) 68% d) 78% e) None of the above

49. Reena got 76 marks in Hindi, 48 marks in Science, 84 marks in Mathematics, 38 marks in Social Science and 72 marks in English. The maximum marks of each subject was 100. How much over all percentage of marks she got?

    a) 63.8% b) 62.6% c) 63.6% d) 64.8% e) None of the above

50. Sahil got 45 marks in Hindi, 47 marks in Science, 49 marks in Mathematics, 45 marks in Social Science and 32 marks in English. The maximum marks of each subject is 50. How much overall percentage of marks did he get?

    a) 75% b) 56.4% c) 78% d) 87.2% e) None of the above

51. A student was asked to measure the length and breadth of a rectangle. By mistake, he measured the length 20% less and the breadth 10% more. If its original area is 200 sq cm, then find the area after this measurement?

    a) 176 sq cm b) 206 sq cm c) 226 sq cm d) 316 sq cm e) None of the above

52. A box has 100 blue balls, 50 red balls and 50 black balls. 25% of blue balls and 50% of red balls are taken away. Then, percentage of black balls at present is

    a) 26% b) $33\frac{1}{3}\%$ c) 40% d) 60%

53. A jogger desires to run a certain course in $\frac{1}{4}$ less than he usually takes. By what percent must he increase in average running speed to accomplish the goal?

    a) 50% b) 20% c) 25% d) $33\frac{1}{3}$

54. The prices of two articles are as 3 : 4. If the price of the first article is increased by 10% and that of the second by ₹4, original ratio remains the same. The original price of the second article is

    a) ₹40 b) ₹10 c) ₹30 d) ₹35 e) None of the above

55. A saves 20% of his monthly salary. If his monthly expenditure is ₹6000, then his monthly savings is

    a) ₹1200 b) ₹4800 c) ₹1500 d) ₹1800

56. From 2008 to 2009, the sales of a book decreased by 80%. If the sales in 2010 was the same as in 2008, by what percent did it increase from 2009 to 2010?

    a) 80% b) 100% c) 120% d) 400%

57. In a particular constituency, 75% of voters cast their votes, out of which 2% were rejected. The winning candidate received 75% of the valid votes and bagged a total of 9261 votes. The total number of voters in the constituency is

    a) 14500 b) 18900 c) 16800 d) 24000

58. If increasing 20 by P percentage gives the same result as decreasing 60 by P percentage, what is P percentage of 70?

    a) 50 b) 140 c) 14 d) 35 e) None of the above

59. Last year, there were 610 boys in a school. The number decreased by 20% this year. How many girls are there in the school, if the number of girls is 175% of the total number of boys in the school this year?

    a) 854 b) 848 c) 798 d) 782 e) None of the above

60. Aryan got 350 marks and Vidya scored 76% marks in the same test. If Vidya scored 296 marks more than Aryan, then what were the maximum marks of the test?

    a) 650 b) 900 c) 850 d) 950 e) None of the above

61. A student was awarded certain marks in an examination. However, after re-evaluation, his marks were reduced by 40% of the marks that were originally awarded to him, so that the new score now became 96. How many marks did the student lose after re-evaluation?

    a) 58 b) 68 c) 63 d) 56 e) 64

62. What should come in place of the question mark (?), so that it satisfies equality of the given equation? 32% of 750 = ?

    a) 23% of 600 b) 46% of 207 c) 98% of 250 d) 75% of 320 e) None of the above

Exercise : Higher Skill Level Questions

1. An alloy of gold and silver weights 50g. It contains 80% gold. How much gold should he added to the alloy, so that percentage of gold is increased to 90?

    a) 50 g b) 60 g c) 30 g d) 40 g e) None of the above

2. Veena spends 25% of her monthly income on household expenses. Her annual income is ₹4.32 lakh. What is the total amount that Veena spends on household expenses in 8 months together?

    a) ₹74000 b) ₹71000 c) ₹73000 d) ₹72000 e) None of the above

3. The salary of an employee of a company increases every month by 4%. If his salary in August was ₹6300, then what would be his approximate salary in month of October of the same year?

    a) ₹6552 b) ₹6967 c) ₹6814 d) ₹6627 e) None of the above

4. 1 L of water is added to 5 L of alcohol and water solution containing 40% alcohol strength. The strength of alcohol in the new solution will be

    a) 30% b) $33\frac{1}{3}\%$ c) $33\frac{2}{3}\%$ d) 33%

5. In an examination, 49% students failed in English, 36% students failed in Hindi, while 15% failed in both. If total number of passed students is 450, then how many students did appear in the examination?

    a) 1800 b) 2000 c) 1100 d) 1500 e) None of the above

6. In a test, Rajesh got 112 marks which was 32 more than the passing marks. Sonal got 75% marks which was 70 more than the passing marks. What is the minimum passing percentage of the test?

    a) 35% b) 45% c) 40% d) 48% e) None of the above

7. Bina's monthly income is 90% of Anita's monthly income. The total of both their monthly incomes is Mr. Sen's monthly income. Mr. Sen's annual income is ₹775200. What is Bina's monthly income?

    a) ₹34000 b) ₹36000 c) ₹30600 d) ₹30000 e) None of the above

8. In an examination, it is required to get 296 marks out of aggregate marks to pass. A student gets 222 marks and is declared failed by 10% marks. What are the maximum aggregate marks a student can get?

   a) 830 b) 810 c) 780 d) 740 e) None of the above

9. Sohan spends 23% of an amount of money on an insurance policy, 33% on food, 19% on children's education and 16% on recreation. He deposits the remaining amount of ₹504 in bank. How much total amount does he spend on food and insurance policy together?

   a) ₹3200 b) ₹3126 c) ₹3136 d) ₹3048 e) None of the above

10. Sonali invests 15% of her monthly salary in insurance policies. She spends 55% of her monthly salary in shopping and on household expenses. She saves the remaining amount at ₹12750. What is Sonali is monthly income?

    a) ₹42500 b) ₹38800 c) ₹40000 d) ₹35500 e) None of the above

11. In an office, 40% of the staff is female. 40% of the female and 60% of the male voted for me. The percentage of votes I got was

    a) 24% b) 42% c) 50% d) 52%

12. Ram sells his goods 25% cheaper than Shyam and 25% dearer than Hari. How much percentage is Hari's goods cheaper than Shyam?

    a) 25 b) $33\frac{1}{3}$ c) 40 d) 50

13. The ratio of the number of boys and girls in a school is 3 : 2. If 20% of the boys and 25% of the girls are scholarship holders, then the percentage of the students who do not get the scholarship is

    a) 78% b) 75% c) 60% d) 55%

14. A reduction of 20% in the price of wheat enables a person to buy 3.5 kg more wheat for ₹770. The original price of rice is

    a) ₹55 per kg b) ₹45 per kg c) ₹65 per kg d) ₹37 per kg e) None of the above

15. Fresh grapes contain 80% water, while dry grapes contain 10% water. If the weight of dry grapes is 500 kg, then what is its total weight when it is fresh?

    a) 2350 kg b) 2085 kg c) 2255 kg d) 2250 kg e) None of the above

16. Due to a 25% increase in the price of rice per kilogram, a person is able to purchase 20 kg less for ₹400. What is the increased price of rice per kilogram?

    a) ₹5 b) ₹6 c) ₹10 d) ₹4

17. The price of rice is reduced by 5%. How many kilogrames of rice can now be bought for the money, which was sufficient to buy 50 kg of rice earlier?

    a) 52.63 kg b) 50.40 kg c) 42.30 kg d) 60.50 kg

18. The price of rice decreases by 6.25% and because of this reduction, Vandana is able to buy 1 kg more for ₹120. Find the reduced rate of rice.

    a) ₹7.50 per kg b) ₹9 per kg c) ₹5.50 per kg d) ₹19 per kg

19. Due to an increase of 30% in the price of eggs, 6 eggs less are available for ₹7.80. The present rate of eggs per dozen is

    a) ₹5.50 b) ₹4.68 c) ₹6.49 d) ₹3.58 e) None of the above

20. Wheat is now being sold at ₹27 per kg. During last month, its cost was ₹24 per kg. Find by how much per cent a family reduces its consumption, so as to keep the expenditure fixed.

    a) 10.2% b) 12.1% c) 12.3% d) 11.1%

21. In a class of 40 students and 5 teachers, each student got sweets that are 15% of the total number of students and each teacher got sweets that are 20% of the total number of students. How many sweets were there?

    a) 280 b) 240 c) 320 d) 360 e) None of the above

22. The price of ghee is increased by 32%. Therefore, a family reduces its consumption, so that the increment in price of ghee is only 10%. If consumption of ghee is 10 kg before the increment, then what is the consumption now?

    a) $8\frac{1}{3}$ kg b) $8\frac{3}{4}$ kg c) $8\frac{1}{2}$ kg d) 9 kg

23. The expenses on wheat, meat and vegetables of a family are in the ratio 12 : 17 : 3. The prices of these articles are increased by 20%, 30% and 50% respectively. The total expenses of the family on these articles are increased by

    a) $23\frac{1}{3}\%$ b) $28\frac{1}{8}\%$ c) $27\frac{1}{8}\%$ d) $25\frac{1}{7}\%$ e) None of the above

24. In 1998, ratio of the numbers of students taking examinations in x, y and z states are respectively 3 : 5 : 6. Next year, the numbers of students are increased by 20%, 10% and 20% respectively. If ratio of the numbers of students in states x and z is 1 : 2, then find the number of students who sit to take examination in 1998.

    a) 5000 b) 6000 c) 7500 d) Data is insufficient e) None of the above

25. The tank-full of petrol in Aran's motor-cycle lasts for 10 days. If he starts using 25% more everyday, how many days will the tank-full of petrol last?

    a) 5 b) 6 c) 7 d) 8

26. In an examination out of 480 students, 85% of the girls and 70% of the boys passed. How many boys appeared in the examination, if total pass percentage was 75%?

    a) 370 b) 340 c) 320 d) 360 e) None of the above

27. In a class X of 30 students, 24 passed in first class and in another class Y of 35 students, 28 passed in first class. In which class was the percentage of students first class more?

    a) Class X has more percentage of students getting first class.

    b) Class Y has more percentage of students getting first class

    c) Both classes have equal percentage of students getting first class

    d) None of the above

Basic Skill Level Questions Answer Key													
1	a	11	a	21	b	31	c	41	c	51	a	61	e
2	b	12	c	22	b	32	a	42	c	52	b	62	d
3	a	13	d	23	d	33	d	43	e	53	d		
4	c	14	c	24	c	34	c	44	a	54	a		
5	d	15	b	25	c	35	a	45	c	55	c		
6	b	16	a	26	d	36	c	46	d	56	d		
7	b	17	a	27	e	37	a	47	a	57	c		
8	c	18	c	28	a	38	d	48	d	58	d		
9	d	19	a	29	c	39	b	49	c	59	a		
10	d	20	d	30	a	40	d	50	d	60	c		

Higher Skill Level Questions											
1	a	6	c	11	d	16	a	21	a	26	c
2	d	7	c	12	c	17	a	22	a	27	c
3	c	8	d	13	a	18	a	23	b		
4	b	9	c	14	a	19	b	24	d		
5	d	10	a	15	d	20	d	25	d		